



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2005
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #2

Mathematics Extension 2

General Instructions

- Reading Time – 5 Minutes
- Working time – 2 Hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Section is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 92

- Attempt questions 1 – 6

Examiner: *C.Kourtesis*

Section A
(Start a new answer sheet.)

Question 1. (18 marks)

- | | Marks |
|---|--------------|
| (a) If $z = 1 + 2i$ and $w = 3 - i$ express $z^2 \div \bar{w}$ in the form $a + ib$ (where a and b are real). | 3 |
| (b) Sketch the region in the Argand diagram represented by $ z + i \leq 2$. | 2 |
| (c) Find all pairs of integers a and b that satisfy $(a + ib)^2 = -3 - 4i$. | 3 |
| (d) If $ z - 1 = \operatorname{Re}(z) + 1$ find the locus of z . | 3 |
| (e) If z and w are two complex numbers, prove that
$\overline{z - w} = \bar{z} - \bar{w}.$ | 3 |
| (f) (i) Express each of the complex numbers $z = 2i$ and $w = 1 + i\sqrt{3}$ in modulus-argument form. | 2 |
| (ii) Find the exact value of $\arg(z + w)$. | 2 |

Question 2. (15 marks)

- | | Marks |
|---|--------------|
| (a) Find $\int \frac{dx}{x^2 + 2x + 5}$. | 2 |
| (b) Find $\int \frac{\sin x \, dx}{(2 + 7 \cos x)^5}$ using the substitution $u = 2 + 7 \cos x$. | 2 |
| (c) Use integration by parts to find:
$\int x^3 \ln x \, dx$ | 3 |
| (d) (i) Find real numbers a , b , and c such that
$\frac{6}{x^2(x+3)} = \frac{ax+b}{x^2} + \frac{c}{x+3}.$ | 2 |
| (ii) Find $\int \frac{6}{x^2(x+3)} \, dx$ | 2 |
| (e) Use the substitution $t = \tan \frac{x}{2}$ to evaluate
$\int_0^{\frac{\pi}{3}} \frac{1}{1 - \sin x} \, dx$ | 4 |

Give your answer in simplest exact form.

Section B
(Start a new answer sheet.)

Question 3. (14 marks)

Marks

- (a) Find the equation whose roots are twice those of the equation:

2

$$x^3 - x - 3 = 0.$$

- (b) (i) Prove that if α is a double root of the equation $P(x) = 0$ then $P'(\alpha) = 0$.

2

- (ii) Prove that $P(x) = x^3 - 3ax + b$ has a double root if $b^2 = 4a^3$.

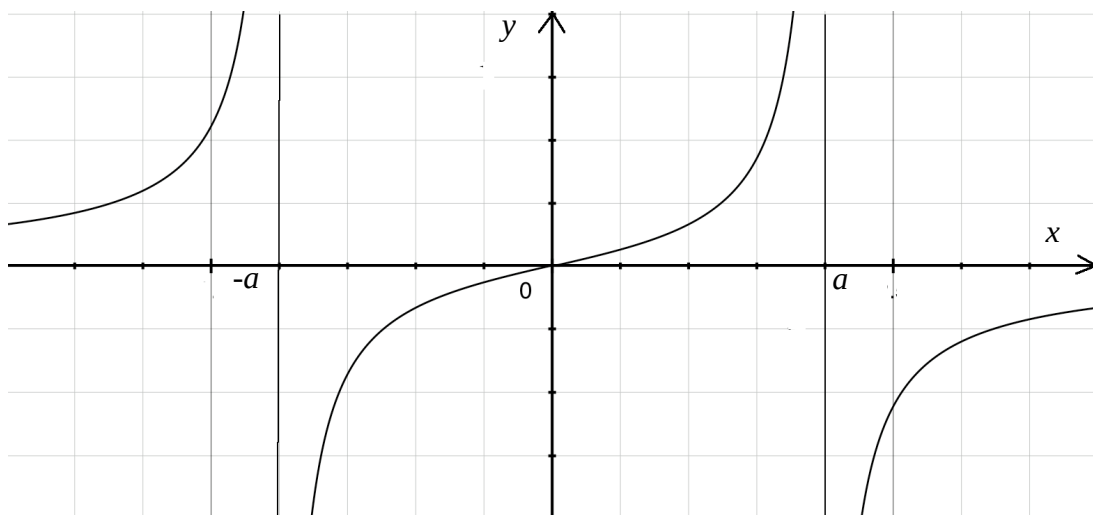
3

- (iii) Hence or otherwise solve the equation

2

$$x^3 + 3x + 2i = 0$$

- (c) The graph of $y = f(x)$ is shown:



Draw sketches of the following curves (on separate diagrams):

- (i) $y = [f(x)]^2$

1

- (ii) $y = f'(x)$

2

- (iii) $y = \frac{1}{f(x)}$

2

Question 4 (15 marks)

(a) Sketch the graphs of:

(i) $y = \sin^2 2x$ for $-2\pi \leq x \leq 2\pi$. 2

(ii) $\sin(x + y) = 0$ 2

(b) (i) Sketch the graph of $y = |x - 1|(x + 1)$. 2

(ii) Hence or otherwise sketch the graph of 2

$$y = \frac{1}{|x - 1|(x + 1)}.$$

(c) Consider the function:

$$f(x) = \frac{e^{-x} - e^x}{e^x + e^{-x}}$$

(i) Show that $f(x)$ is an odd function. 1

(ii) Find the equations of any asymptotes. 2

(iii) Show that $f(x)$ is a decreasing function for all real x . 2

(iv) Sketch the graph of $y = f(x)$. 2

Section C
(Start a new answer booklet)

Question 5 (13 marks)

- (a) Sketch the graph of the function 4

$$y = x^{\frac{1}{3}} + \frac{1}{4}x^{\frac{4}{3}}$$

indicating the nature of any turning points and the co-ordinates of any points of inflexion.

- (b) A rectangle is divided by m lines parallel to one pair of opposite sides and by n lines parallel to the other pair of opposite sides. 2

How many rectangles of any size are formed in the resulting figure?
(Leave your answer in unsimplified form.)

- (c) Given that $z_1 = f(z) = \frac{z+i}{z-i}$, show that $f(z_1) = \left(\frac{z+1}{z-1}\right) \cdot i$ 3

- (d) If a and b are two roots of the equation $x^3 + 4x - 2 = 0$, show that ab is a root of the equation $x^3 - 4x^2 - 4 = 0$. 4

Question 6 (17 marks)

(a) Consider the function

$$f(x) = x + \log_e(1-x)$$

(i) Sketch the graph of $y = f(x)$ showing all essential features. 3

(ii) Hence show that $x \leq -\log_e(1-x)$ for all $x < 1$. 1

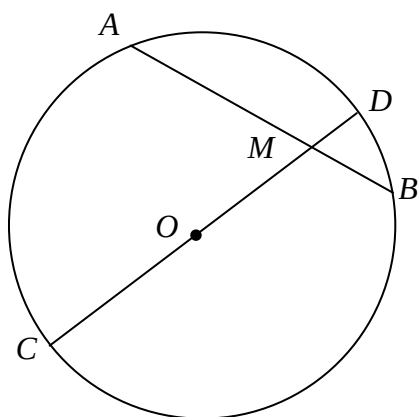
(b) (i) Show that $\int_0^{\frac{\pi}{2}} x \cos x \, dx = \frac{\pi}{2} - 1$. 2

(ii) If $u_n = \int_0^{\frac{\pi}{2}} x \cos^n x \, dx$ and n is a positive integer greater than 1, prove that 4

$$u_n = \left(\frac{n-1}{n} \right) u_{n-2} - \frac{1}{n^2}.$$

(iii) Deduce that $u_5 = \frac{4\pi}{15} - \frac{149}{225}$. 2

(c)



A chord AB and a diameter CD of a circle centre O intersect at a point M within the circle.
(M is not the centre.)

(i) Show that $(CM + MD)^2 > (AM + MB)^2$ 2

(ii) Deduce that $(CM - MD)^2 > (AM - MB)^2$ 3

This is the end of the paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, x > 0$



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Sample Solutions

Section	Marker
A	PSP
B	PRB
C	DMH

Section A

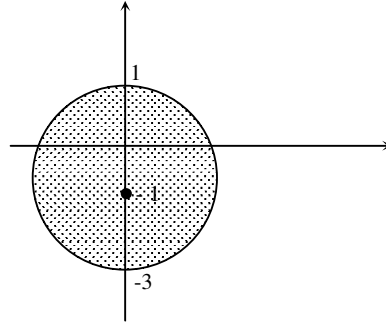
Question 1

(a) $z = 1 + 2i, w = 3 - i$

$$\therefore z^2 = 1 - 4 + 4i = -3 + 4i, \bar{w} = 3 + i$$

$$\therefore \frac{z^2}{\bar{w}} = \frac{-3 + 4i}{3 + i} \times \frac{3 - i}{3 - i} = \frac{-5 + 15i}{10} = -\frac{1}{2} + \frac{3}{2}i$$

(b) The interior of a circle centred at $z = -i$ and radius of 2 units.



(c) $z = a + ib$

$$z^2 = (a + ib)^2 = -3 - 4i$$

$$\therefore a^2 - b^2 = -3,$$

$$2ab = -4 \Rightarrow ab = -2$$

$$|z^2| = 5 \Rightarrow |z|^2 = 5 \Rightarrow a^2 + b^2 = 5$$

$$\left. \begin{array}{l} a^2 + b^2 = 5 \\ a^2 - b^2 = -3 \end{array} \right\} +$$

$$\therefore 2a^2 = 2 \Rightarrow a = \pm 1$$

$$\therefore 2b^2 = 8 \Rightarrow b = \pm 2$$

$$\therefore ab < 0 \Rightarrow z = \pm(1 - 2i)$$

$$\therefore (1, -2) \& (-1, 2)$$

(d) $|z - 1| = \operatorname{Re}(z) + 1$

$$z = x + iy$$

$$\therefore |(x - 1) + iy| = x + 1 \Rightarrow x \geq -1$$

$$\therefore \sqrt{(x - 1)^2 + y^2} = x + 1$$

$$\therefore (x - 1)^2 + y^2 = (x + 1)^2$$

$$\therefore y^2 = (x + 1)^2 - (x - 1)^2 = (x + 1 - x + 1)(x + 1 + x - 1)$$

$$\therefore y^2 = (2)(2x) = 4x$$

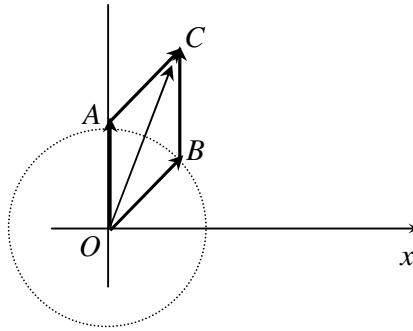
$$\therefore y^2 = 4x$$

$$\begin{aligned}
 \text{(e)} \quad & z = z_1 + iz_2, w = w_1 + iw_2 \\
 & \therefore z - w = (z_1 - w_1) + i(z_2 - w_2) \\
 & \therefore \overline{z - w} = (z_1 - w_1) - i(z_2 - w_2) \\
 & \overline{z} - \overline{w} = (z_1 - iz_2) - (w_1 - iw_2) = (z_1 - w_1) - i(z_2 - w_2) \\
 & \therefore \overline{z - w} = \overline{z} - \overline{w}
 \end{aligned}$$

$$\begin{aligned}
 \text{(f)} \quad \text{(i)} \quad & |2i| = 2, \arg(i) = \frac{\pi}{2} \Rightarrow z = 2\text{cis} \frac{\pi}{2} \\
 & |1 + i\sqrt{3}| = 2, \arg(1 + i\sqrt{3}) = \frac{\pi}{3} \Rightarrow w = 2\text{cis} \frac{\pi}{3}
 \end{aligned}$$

(ii) If C corresponds to the complex number $z + w$ then $OACB$ is a rhombus.

$$\text{So } \angle COA = \frac{1}{2} \angle AOB = \frac{\pi}{12} \quad [\text{properties of rhombi}]$$



$$\begin{aligned}
 \arg(z + w) &= \angle COx \\
 &= \angle BOx + \frac{1}{2}(\angle AOB) \\
 &= \frac{\pi}{3} + \frac{1}{2}\left(\frac{\pi}{6}\right) \\
 &= \frac{5\pi}{12}
 \end{aligned}$$

Question 2

$$\begin{aligned} \text{(a)} \quad \int \frac{dx}{x^2 + 2x + 5} &= \int \frac{dx}{(x+1)^2 + 4} \\ &= \frac{1}{2} \tan\left(\frac{x+1}{2}\right) + c \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad u &= 2 + 7 \cos x \\ du &= -7 \sin x dx \\ \int \frac{\sin x dx}{(2 + 7 \cos x)^5} &= -\frac{1}{7} \int \frac{-7 \sin x dx}{(2 + 7 \cos x)^5} \\ &= -\frac{1}{7} \int \frac{du}{u^5} \\ &= -\frac{1}{7} \int u^{-5} du \\ &= -\frac{1}{7} \left[-\frac{1}{4} u^{-4} \right] + c \\ &= \frac{1}{28(2 + 7 \cos x)^4} + c \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad \int \underbrace{x^3}_{f'} \underbrace{\ln x}_g dx &= fg - \int fg' dx \\ &= \frac{1}{4} x^4 \ln x - \int \left(\frac{1}{4} x^4 \times \frac{1}{x} \right) dx \\ &= \frac{1}{4} x^4 \ln x - \frac{1}{4} \int x^3 dx \\ &= \frac{1}{4} x^4 \ln x - \frac{1}{16} x^4 + c \end{aligned}$$

$$\begin{aligned} \text{(d)} \quad \text{(i)} \quad \frac{6}{x^2(x+3)} &= \frac{ax+b}{x^2} + \frac{c}{x+3} \\ \therefore 6 &= (ax+b)(x+3) + cx^2 \\ \text{sub } x &= -3: 6 = c(9) \Rightarrow c = \frac{2}{3} \\ a+c &= 0 \Rightarrow a = -\frac{2}{3} \quad \left[\text{coefficient of } x^2 \right] \\ 3b &= 6 \Rightarrow b = 2 \quad \left[\text{constant term} \right] \\ \therefore a &= -\frac{2}{3}, b = 2, c = \frac{2}{3} \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \frac{6}{x^2(x+3)} &= \frac{-\frac{2}{3}x+2}{x^2} + \frac{\frac{2}{3}}{x+3} \\
 &= -\frac{2}{3} \left(\frac{x-3}{x^2} - \frac{1}{x+3} \right) \\
 &= -\frac{2}{3} \left(\frac{1}{x} - 3x^{-2} - \frac{1}{x+3} \right)
 \end{aligned}$$

$$\begin{aligned}
 \int \frac{6}{x^2(x+3)} dx &= -\frac{2}{3} \int \left(\frac{1}{x} - 3x^{-2} - \frac{1}{x+3} \right) dx \\
 &= -\frac{2}{3} \left[\ln x + 3x^{-1} - \ln(x+3) \right] + c \\
 &= -\frac{2}{3} \ln \left(\frac{x}{x+3} \right) - \frac{2}{x} + c \\
 &= \frac{2}{3} \ln \left(\frac{x+3}{x} \right) - \frac{2}{x} + c
 \end{aligned}$$

$$\text{(e)} \quad t = \tan \left(\frac{x}{2} \right) \Rightarrow x = 2 \tan^{-1}(t)$$

$$\therefore dx = \frac{2}{1+t^2} dt$$

$$x = 0 \Rightarrow t = 0$$

$$x = \frac{\pi}{3} \Rightarrow t = \frac{1}{\sqrt{3}}$$

$$\sin x = \frac{2t}{1+t^2} \Rightarrow 1 - \sin x = \frac{1+t^2-2t}{1+t^2} = \frac{(t-1)^2}{1+t^2}$$

$$\begin{aligned}
\int_0^{\frac{\pi}{3}} \frac{1}{1-\sin x} dx &= \int_0^{\frac{1}{\sqrt{3}}} \left(\frac{1}{\frac{(t-1)^2}{1+t^2}} \right) \times \frac{2dt}{1+t^2} = \int_0^{\frac{1}{\sqrt{3}}} \frac{2dt}{(t-1)^2} \\
&= 2 \int_0^{\frac{1}{\sqrt{3}}} (t-1)^{-2} dt \\
&= -2 \left[(t-1)^{-1} \right]_0^{\frac{1}{\sqrt{3}}} \\
&= -2 \left[\frac{1}{t-1} \right]_0^{\frac{1}{\sqrt{3}}} \\
&= -2 \left[\frac{1}{\frac{1}{\sqrt{3}}-1} - (-1) \right] \\
&= -2 \left[\frac{\sqrt{3}}{1-\sqrt{3}} + 1 \right] \\
&= \frac{2\sqrt{3}}{\sqrt{3}-1} - 2 \\
&= \frac{2}{\sqrt{3}-1} \\
&= \sqrt{3}+1
\end{aligned}$$

Section B

QUESTION 3

a) Given $x^3 - x - 3 = 0$ — (A)

Form a new equation where $x = 2x \Rightarrow x = \frac{x}{2}$ in (A)

$$\left(\frac{x}{2}\right)^3 - \frac{x}{2} - 3 = 0$$

$$\frac{x^3}{8} - \frac{x}{2} - 3 = 0$$

$$\boxed{x^3 - 4x - 24 = 0}$$

b) (i) Let $P(x) = (x - \alpha)^2 \cdot Q(x)$.

$$\therefore P'(x) = 2(x - \alpha) \cdot Q(x) + (x - \alpha)^2 \cdot Q'(x)$$

$$\therefore P'(\alpha) = 2(\alpha - \alpha) \cdot Q(\alpha) + (\alpha - \alpha)^2 \cdot Q'(\alpha)$$

$$= 2 \times 0 \times Q(\alpha) + 0 \times Q'(\alpha)$$

$$= 0$$

(ii) Given $P(x) = x^3 - 3ax + b$ — (A)

then $P'(x) = 3x^2 - 3a$ — (B)

If $P(x) = 0$ has a double root, say α .
then $P(\alpha) = P'(\alpha) = 0$.

From (B) If $P'(\alpha) = 0$ $3\alpha^2 - 3a = 0$

$$\therefore \alpha^2 = a$$

In (A) $\alpha^3 - 3\alpha a + b = 0$

$$\alpha(\alpha^2 - 3a) = -b$$

$$\alpha(\alpha - 3a) = -b$$

$$\alpha^2 - 3a = -\frac{b}{\alpha}$$

$$\alpha^2 + 4a = \frac{b^2}{\alpha^2}$$

$$\therefore \alpha^2 + 4a = \frac{b^2}{\alpha^2}$$

$$\boxed{4a^3 = b^2}$$

(iii) Consider $x^3 + 3x + 2i = 0$

now $a = -1 \Rightarrow 4a^3 = -4$

and $b = 2i \Rightarrow b^2 = -4$

$\therefore$ a double root exists.

Consider $3x^2 = 3a$

$x^2 = -1$

$x = \pm i$.

now $P(i) = -i + 3i + 2i \neq 0$

$\therefore P(-i) = i + 3i + 2i = 0$

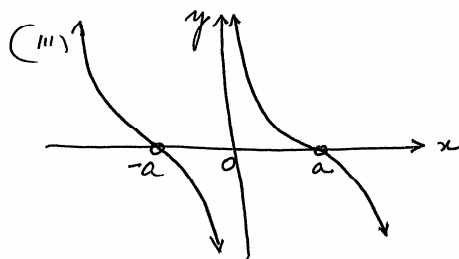
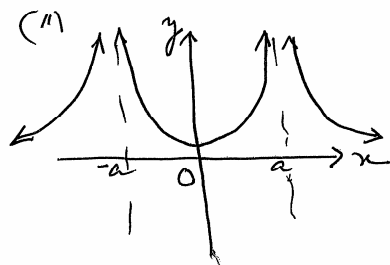
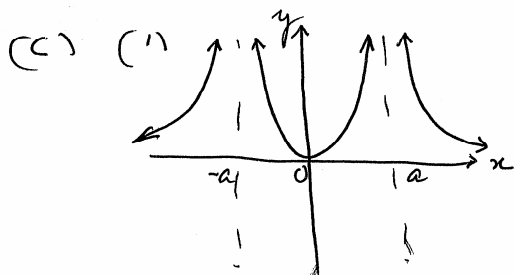
$\therefore -i$ is the double root.

now $S_1 = \sum \alpha = 0$

$\therefore -i + i + \beta = 0$

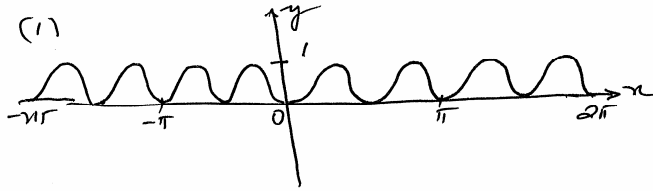
$\beta = 2i$

$\therefore$ roots are $\boxed{-i, -i, 2i}$.

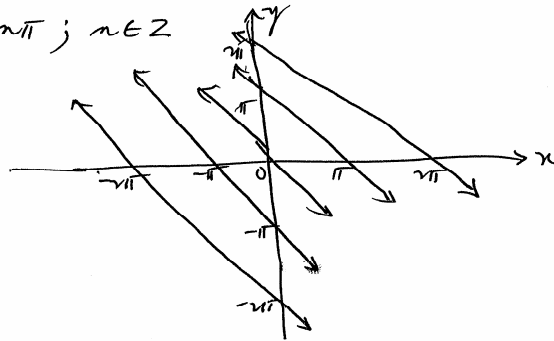


QUESTION 4

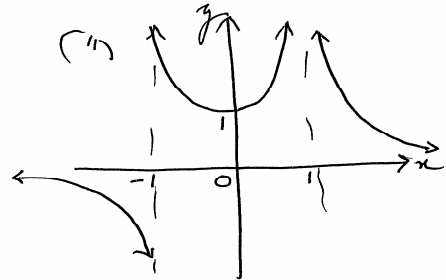
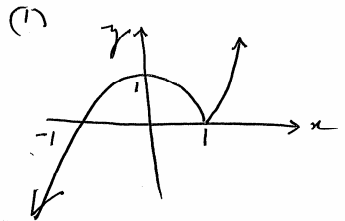
(a) (i)



(ii) $\sin(x+y) = 0$
 $x+y = n\pi; n \in \mathbb{Z}$



(b)



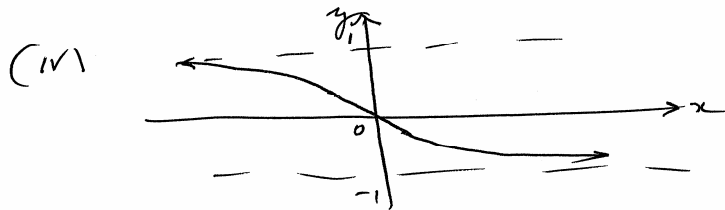
(c) (i) $f(x) = \frac{e^{-x} - e^x}{e^x + e^{-x}}$
 $f(-x) = \frac{e^x - e^{-x}}{e^{-x} + e^x} = -\frac{(e^{-x} - e^x)}{e^x + e^{-x}} = -f(x) \therefore \text{ODD!!}$

(ii) For vertical asymptotes
 $e^x + e^{-x} = 0$
 $e^{2x} + 1 = 0$
 $e^{2x} = -1$ NO!!

$f(x) = \frac{e^{-x} - 1}{1 + e^{-x}} = \frac{1 - e^x}{e^x + 1}$
 as $x \rightarrow \infty$ $f(x) \rightarrow \frac{-1}{1} = -1$
 as $x \rightarrow -\infty$ $f(x) \rightarrow \frac{1}{1} = 1$

$\therefore$ horizontal asymptote $y = 1$ as $x \rightarrow -\infty$
 $y = -1$ as $x \rightarrow \infty$.

$$\begin{aligned}
 \text{(iii)} \quad f'(x) &= \frac{(e^x + e^{-x})(-e^{-x} - e^x) - (e^{-x} - e^x)(e^x - e^{-x})}{(e^x + e^{-x})^2} \\
 &= \frac{-1 - e^{2x} - e^{-2x} - 1 + e^{2x} + e^{-2x} - 1}{(e^x + e^{-x})^2} \\
 &= \frac{-4}{(e^x + e^{-x})^2} \\
 &< 0 \quad \therefore \text{decreasing.}
 \end{aligned}$$



Section C

5. (a) Sketch the graph of the function

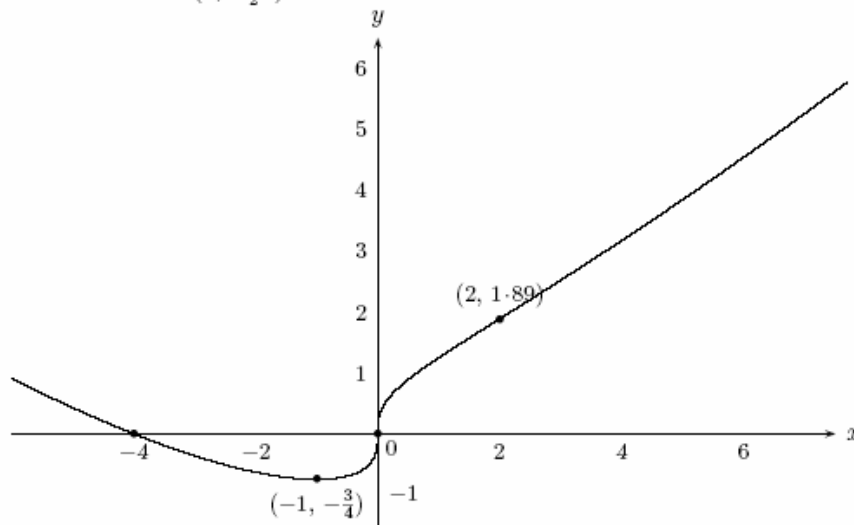
$$y = x^{\frac{1}{3}} + \frac{1}{4}x^{\frac{4}{3}}$$

indicating the nature of any turning points and the nature of any points of inflexion.

Solution:

$$\begin{aligned} y &= \frac{\sqrt[3]{x}}{4}(4+x), \\ &= 0 \text{ when } x = 0, -4. \\ y' &= \frac{1}{3}x^{-\frac{2}{3}} + \frac{1}{3}x^{\frac{1}{3}}, \\ &= \frac{(1+x)}{3\sqrt[3]{x^2}}, \\ &= 0 \text{ when } x = -1, \\ &= \text{undefined when } x = 0. \\ y'' &= -\frac{2x^{-\frac{5}{3}}}{9} + \frac{x^{-\frac{2}{3}}}{9}, \\ &= \frac{x-2}{9x\sqrt[3]{x^2}}, \\ &= \frac{1}{48} \text{ when } x = 8, \\ &= 0 \text{ when } x = 2, \\ &= -\frac{1}{3} \text{ when } x = 1, \\ &= \frac{1}{3} \text{ when } x = -1. \end{aligned}$$

$\therefore$ Minimum at $(-1, -\frac{3}{4})$,
vertical inflexion at $(0, 0)$,
inflexion at $(2, \frac{3\sqrt[3]{2}}{2})$.



- (b) A rectangle is divided by m lines parallel to one pair of opposite sides and by n lines parallel to the other pair of opposite sides.

How many rectangles of any size are formed in the resulting figure?
(Leave your answer in unsimplified form.)

Solution: $\binom{m+2}{2} \times \binom{n+2}{2}$
or ${}^{m+2}C_2 \times {}^{n+2}C_2$.

- (c) Given that $z_1 = f(z) = \frac{z+i}{z-i}$, show that $f(z_1) = \left(\frac{z+1}{z-1}\right) \cdot i$

Solution:
$$\begin{aligned} f(z_1) &= \frac{\frac{z+i}{z-i} + i}{\frac{z+i}{z-i} - i}, \\ &= \frac{z+i+i(z-i)}{z+i-i(z-i)}, \\ &= \frac{z+i+iz+1}{z+i-iz-1}, \\ &= \frac{z+1+i(z+1)}{z-1-i(z-1)}, \\ &= \frac{(z+1)(1+i)}{(z-1)(1-i)} \times \frac{1+i}{1+i}, \\ &= \left(\frac{z+1}{z-1}\right) \left(\frac{1+2i-1}{1+1}\right), \\ &= \left(\frac{z+1}{z-1}\right) \cdot i \end{aligned}$$

- (d) If a and b are two roots of the equation $x^3 + 4x - 2 = 0$, show that ab is a root of the equation $x^3 - 4x^2 - 4 = 0$.

Solution: Let c be the third root of $x^3 + 4x - 2 = 0$,

$$\text{then, } a + b + c = 0 \dots \boxed{1}$$

$$ab + bc + ca = 4 \dots \boxed{2}$$

$$abc = 2 \dots \boxed{3}$$

Method 1: From $\boxed{1}$ $c = -(a+b) \dots \boxed{4}$

$$\text{From } \boxed{3} \quad c = \frac{2}{ab} \dots \boxed{5}$$

$$\begin{aligned} \text{Sub. } \boxed{5} \text{ in } \boxed{2}, \quad ab + \frac{2b}{ab} + \frac{2a}{ab} &= 4 \\ a^2b^2 + 2(a+b) &= 4ab \\ a^2b^2 - 2c &= 4ab \end{aligned}$$

$$a^2b^2 - \frac{4}{ab} - 4ab = 0$$

$$(ab)^3 - 4(ab)^2 - 4 = 0$$

i.e., ab is a root of $x^3 - 4x^2 - 4 = 0$.

Method 2: Consider the equation with roots ab, bc, ca .

Sum of roots one at a time is $ab + bc + ca = 4$.

Sum of roots two at a time is $ab^2c + bc^2a + ca^2b = abc(a + b + c)$
 $= 0$.

Product of roots is $ab^2c^2a = (abc)^2$,
 $= 4$.

$\therefore$ The equation is $x^3 - 4x^2 - 4 = 0$.

Method 3: $ab = \frac{2}{c}$ (from [3] above).

Substitute into $x^3 - 4x^2 - 4 = 0$,

$$\begin{aligned}\text{L.H.S.} &= \frac{b}{c^3} - \frac{16}{c^2} - 4, \\ &= \frac{-4c^3 - 16c + 8}{c^3}, \\ &= \frac{-4}{c^3}(c^3 + 4c - 2)\end{aligned}$$

But c is a root of $x^3 + 4x - 2 = 0$,

$$\begin{aligned}\text{L.H.S.} &= 0, \\ &= \text{R.H.S.}\end{aligned}$$

$\therefore ab$ is a root of $x^3 - 4x^2 - 4 = 0$.

6. (a) Consider the function

$$f(x) = x + \log_e(1 - x)$$

i. Sketch the graph of $y = f(x)$, showing all essential features.

Solution: $f(x) = x + \ln(1 - x), \quad x < 1$

$$f(0) = 0.$$

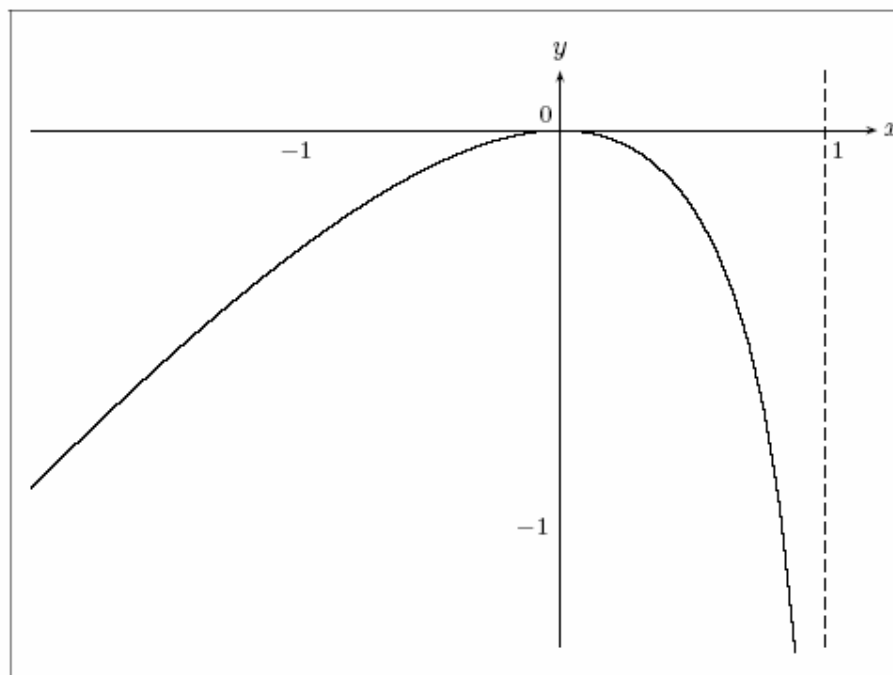
$$\begin{aligned}f'(x) &= 1 - \frac{1}{1-x}, \\ &= \frac{-x}{1-x}, \\ &> 0 \text{ for } x < 0, \\ &= 0 \text{ for } x = 0, \\ &< 0 \text{ for } 0 < x < 1.\end{aligned}$$

$$f'(1) = \text{undefined}.$$

$$\begin{aligned}f''(x) &= \frac{-1}{(1-x)^2}, \\ &< 0 \text{ for all } x.\end{aligned}$$

Hence maximum at $(0, 0)$, no inflexions.

Vertical asymptote at $x = 1$.



- ii. Hence show that $x \leq \log_e(1-x)$ for all $x \leq 1$.

Solution: First interpretation—

Question should have read:

Hence show that $x \leq -\log_e(1-x)$ for all $x \leq 1$.

i.e., $x + \ln(1-x) \leq 0$.

And it is clear from the sketch above that $f(x) \leq 0$, $x \leq 1$.

.....
Second interpretation—

Let the question stand, *i.e.*, $x - \ln(1-x) \leq 0$.

Now consider $y = f(x) = x - \ln(1-x)$,

$$f(0) = 0.$$

$$f'(x) = 1 + \frac{1}{1-x},$$

$$> 0 \text{ for } x < 1,$$

$$= \text{undefined when } x = 1.$$

The curve is monotonic increasing and passes through $(0, 0)$. Hence the statement is false.

- (b) i. Show that $\int_0^{\frac{\pi}{2}} x \cos x \, dx = \frac{\pi}{2} - 1$.

Solution:	$I = \int_0^{\frac{\pi}{2}} x \cos x \, dx,$	$u = x$	$v' = \cos x$
	$= x \sin x \Big _0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x \, dx,$	$u' = 1$	$v = \sin x$
	$= \frac{\pi}{2} - 0 - [-\cos x]_0^{\frac{\pi}{2}},$		
	$= \frac{\pi}{2} - 1.$		

- ii. If $u_n = \int_0^{\frac{\pi}{2}} x \cos^n x \, dx$ and n is a positive integer, prove that

$$u_n = \left(\frac{n-1}{n} \right) u_{n-2} - \frac{1}{n^2}.$$

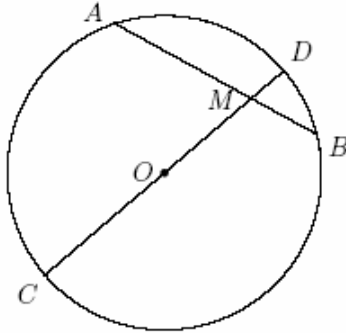
Solution:	
$U_n = \int_0^{\frac{\pi}{2}} x \cos^n x \, dx,$	$u = \cos^{n-1} x$
$= [x \sin x \cos^{n-1} x + \cos^n x]_0^{\frac{\pi}{2}}$	$v' = x \cos x$
$+ (n-1) \int_0^{\frac{\pi}{2}} x \sin^2 x \cos^{n-2} x \, dx$	$u' = (n-1)(-\sin x) \cos^{n-2} x$
$+ (n-1) \int_0^{\frac{\pi}{2}} \sin x \cos^{n-1} x \, dx,$	$v = x \sin x + \cos x$
$= [x \sin x \cos^{n-1} x + \cos^n x]_0^{\frac{\pi}{2}}$	
$+ (n-1) \int_0^{\frac{\pi}{2}} x \cos^{n-2} x \, dx$	
$- (n-1) \int_0^{\frac{\pi}{2}} x \cos^n x \, dx$	
$- (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-1} x \, d \cos x,$	
$= [x \sin x \cos^{n-1} x + \cos^n x$	
$\quad - \frac{(n-1) \cos^n x}{n}]_0^{\frac{\pi}{2}}$	
$+ (n-1)U_{n-2} - (n-1)U_n.$	
$nU_n = \left[x \sin x \cos^{n-1} x + \frac{\cos^n x}{n} \right]_0^{\frac{\pi}{2}}$	
$+ (n-1)U_{n-2},$	
$= -\frac{1}{n} + (n-1)U_{n-2}.$	
$\therefore U_n = \frac{n-1}{n} U_{n-2} - \frac{1}{n^2}.$	

iii. Deduce that $u_5 = \frac{4\pi}{15} - \frac{149}{225}$.

Solution:

$$\begin{aligned}
 U_5 &= \frac{4}{5}U_3 - \frac{1}{25}, \\
 U_3 &= \frac{2}{3}U_1 - \frac{1}{9}, \\
 U_1 &= \frac{\pi}{2} - 1. \\
 \therefore U_5 &= \frac{4}{5} \left(\frac{2}{3} \left\{ \frac{\pi}{2} - 1 \right\} - \frac{1}{9} \right) - \frac{1}{25}, \\
 &= \frac{4\pi}{15} - \frac{8}{15} - \frac{4}{45} - \frac{1}{25}, \\
 &= \frac{4\pi}{15} - \frac{149}{225}.
 \end{aligned}$$

(c)



A chord AB and a diameter CD of a circle centre O intersect at a point M within the circle.

(M is not the centre.)

i. Show that $(CM + MD)^2 > (AM + MB)^2$.

Solution: $CM + MD = CD$, a diameter.
 $AM + MB = AB$, a chord which is not a diameter.
 $CD > AB$, as a diameter is the longest chord in a circle.
 $\therefore CD^2 > AB^2$.
 So $(CM + MD)^2 > (AM + MB)^2$.

ii. Deduce that $(CM - MD)^2 > (AM - MB)^2$.

Solution:

$$\begin{aligned}
 (CM + MD)^2 &> (AM + MB)^2, \\
 CM^2 + 2CM.MD + MD^2 &> AM^2 + 2AM.MB + MB^2.
 \end{aligned}$$

Now $CM.MD = AM.MB$ (intersecting chord theorem),
 so subtract $4CM.MD [= 4AM.MB]$ from both sides,
 $CM^2 - 2CM.MD + MD^2 > AM^2 - 2AM.MB + MB^2$.
 i.e., $(CM - MD)^2 > (AM - MB)^2$.